

Lec 8

Thursday, 29 May 2025 14.17

Opgave 8.1

Analysér og bestem de kritiske punkter for det elektriske kresløb i Eksempel 2, sektion 4.1 (eksempel med elektrisk netværk der blev gennemgået i forbindelse med undervisningen), dvs., bestem de punkter der opfylder

$$\frac{di_2/dt}{di_1/dt} = 0$$

- Hvor ligger det/de kritiske punkter og hvilken type er de?
- Lav et faseplot i området omkring de kritiske punkter, svarende til Fig 80.b, s. 134 i Kreyszig.

Electrical Network

Find the currents $i_1(t)$ and $i_2(t)$ in the network in Fig. 79. Assume all currents and charges to be zero at $t = 0$, the instant when the switch is closed.

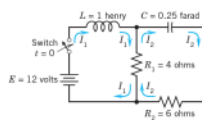


Fig. 79. Electrical network in Example 2

Solution. *Step 1. Setting up the mathematical model.* The model of this network is obtained from Kirchhoff's Voltage Law, as in Sec. 2.9 (where we considered single circuits). Let $i_1(t)$ and $i_2(t)$ be the currents

s of ODEs as Models in Engineering Applications

133

in the left and right loops, respectively. In the left loop, the voltage drops are $L\dot{i}_1 = i_1'$ [V] over the inductor and $R_1(i_1 - i_2) = 4(i_1 - i_2)$ [V] over the resistor, the difference because i_1 and i_2 flow through the resistor in opposite directions. By Kirchhoff's Voltage Law the sum of these drops equals the voltage of the battery; that is, $i_1' + 4(i_1 - i_2) = 12$, hence

$$(4a) \quad i_1' = -4i_1 + 4i_2 + 12.$$

In the right loop, the voltage drops are $R_2i_2 = 6i_2$ [V] and $R_1(i_2 - i_1) = 4(i_2 - i_1)$ [V] over the resistors and $(1/C)\int i_2 dt = 4\int i_2 dt$ [V] over the capacitor, and their sum is zero,

$$6i_2 + 4(i_2 - i_1) + 4 \int i_2 dt = 0 \quad \text{or} \quad 10i_2 - 4i_1 + 4 \int i_2 dt = 0.$$

Division by 10 and differentiation gives $i_2' - 0.4i_1' + 0.4i_2 = 0$.

To simplify the solution process, we first get rid of $0.4i_1'$, which by (4a) equals $0.4(-4i_1 + 4i_2 + 12)$. Substitution into the present ODE gives

$$i_2' = 0.4i_1' - 0.4i_2 = 0.4(-4i_1 + 4i_2 + 12) - 0.4i_2$$

and by simplification

$$(4b) \quad i_2' = -1.6i_1 + 1.2i_2 + 4.8.$$

In matrix form, (4) is (we write $\mathbf{J}$ since $\mathbf{I}$ is the unit matrix)

$$(5) \quad \mathbf{J}' = \mathbf{A}\mathbf{J} + \mathbf{g}, \quad \text{where} \quad \mathbf{J} = \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} -4.0 & 4.0 \\ -1.6 & 1.2 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 12.0 \\ 4.8 \end{bmatrix}$$

Step 2. Solving (5). Because of the vector $\mathbf{g}$ this is a nonhomogeneous system, and we try to proceed as for a single ODE, solving first the homogeneous system $\mathbf{J}' = \mathbf{A}\mathbf{J}$ (thus $\mathbf{J}' - \mathbf{A}\mathbf{J} = \mathbf{0}$) by substituting $\mathbf{J} = \mathbf{x}e^{\lambda t}$. This gives

$$\mathbf{J}' = \lambda \mathbf{x}e^{\lambda t} = \mathbf{A}\mathbf{x}e^{\lambda t}, \quad \text{hence} \quad \mathbf{A}\mathbf{x} = \lambda \mathbf{x}.$$

Hence, to obtain a nontrivial solution, we again need the eigenvalues and eigenvectors. For the present matrix $\mathbf{A}$ they are derived in Example 1 in Sec. 4.0:

$$\lambda_1 = -2, \quad \mathbf{x}^{(1)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \lambda_2 = -0.8, \quad \mathbf{x}^{(2)} = \begin{bmatrix} 1 \\ 0.8 \end{bmatrix}.$$

Hence a "general solution" of the homogeneous system is

$$\mathbf{J}_h = c_1 \mathbf{x}^{(1)} e^{-2t} + c_2 \mathbf{x}^{(2)} e^{-0.8t}.$$

For a particular solution of the nonhomogeneous system (5), since $\mathbf{g}$ is constant, we try a constant column vector $\mathbf{J}_p = \mathbf{a}$ with components a_1, a_2 . Then $\mathbf{J}_p' = \mathbf{0}$, and substitution into (5) gives $\mathbf{A}\mathbf{a} + \mathbf{g} = \mathbf{0}$; in components,

$$-4.0a_1 + 4.0a_2 + 12.0 = 0$$

$$-1.6a_1 + 1.2a_2 + 4.8 = 0.$$

The solution is $a_1 = 3, a_2 = 0$; thus $\mathbf{a} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$.

$$(6) \quad \mathbf{J} = \mathbf{J}_h + \mathbf{J}_p = c_1 \mathbf{x}^{(1)} e^{-2t} + c_2 \mathbf{x}^{(2)} e^{-0.8t} + \mathbf{a}$$

in components,

$$i_1 = 2c_1 e^{-2t} + c_2 e^{-0.8t} + 3$$

$$i_2 = c_1 e^{-2t} + 0.8c_2 e^{-0.8t}.$$

CHAP. 4 Systems of ODEs. Phase Plane. Qualitative Methods

The initial conditions give

$$i_1(0) = 2c_1 + c_2 + 3 = 0$$

$$i_2(0) = c_1 + 0.8c_2 = 0.$$

Hence $c_1 = -4$ and $c_2 = 5$. As the solution of our problem we thus obtain

$$(7) \quad \mathbf{J} = -4\mathbf{x}^{(1)} e^{-2t} + 5\mathbf{x}^{(2)} e^{-0.8t} + \mathbf{a}.$$

The solution is $a_1 = 3, a_2 = 0$; thus $\mathbf{a} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ Hence **Kritisk Punkt**

$$(6) \quad \mathbf{J} = \mathbf{J}_h + \mathbf{J}_p = c_1 x^{(1)} e^{-2t} + c_2 x^{(2)} e^{-0.8t} + \mathbf{a};$$

in components,

$$I_1 = 2c_1 e^{-2t} + c_2 e^{-0.8t} + 3$$

$$I_2 = c_1 e^{-2t} + 0.8c_2 e^{-0.8t}.$$

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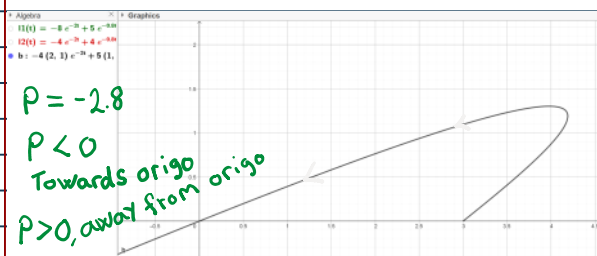
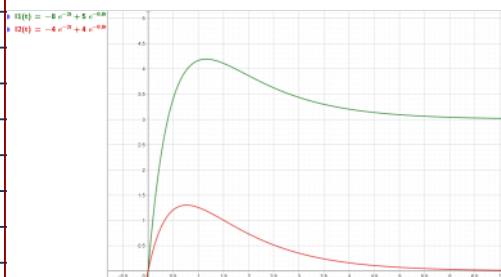
$$(7) \quad \mathbf{J} = -4x^{(1)}e^{-2t} + 5x^{(2)}e^{-0.8t} + \mathbf{a}.$$

In components (Fig. 80b),

$$I_1 = -8e^{-2t} + 5e^{-0.8t} + 3$$

$$I_2 = -4e^{-2t} + 4e^{-0.8t}.$$

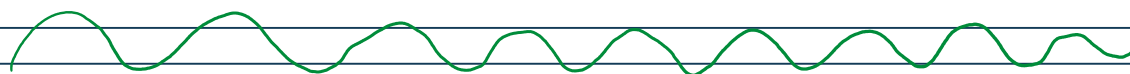
Now comes an important idea, on which we shall elaborate further, beginning in Sec. 4.3. Figure 80a shows $I_1(t)$ and $I_2(t)$ as two separate curves. Figure 80b shows these two currents as a single curve $[I_1(t), I_2(t)]$ in the I_1I_2 -plane. This is a parametric representation with time t as the parameter. It is often important to know in which sense such a curve is traced. This can be indicated by an arrow in the sense of increasing t , as is shown. The I_1I_2 -plane is called the **phase plane** of our system (5), and the curve in Fig. 80b is called a **trajectory**. We shall see that such "phase plane representations" are far more important than graphs as in Fig. 80a because they will give a much better qualitative overall impression of the general behavior of whole families of solutions, not merely of one solution as in the present case.



$$P = -2.8$$

$$q = 1.6$$

$$\Delta = 1.44$$



Opgave 8.2

Betrakt følgende ulineære 2. ordens differentiaalligning:

$$y'' - 9y + y^3 = 0$$

- Opstil det ækvivalente 1. ordens (ulineære) ligningssystem og bestem de kritiske punkter. *Hint: ligningssystemet er en "stak" af ligninger $y'_i = f_i(y_1, \dots, y_n, t)$, hvor f_i generelt er en ulineær funktion - se slides fra lektion 7: "A first order linear system" vs. "A first order linear system".*
- Opstil det generelle *lineariserede* ligningssystem for det kritiske punkt $\tilde{\mathbf{y}} = (\tilde{y}_1, \tilde{y}_2)$; følg proceduren "Linearization (around stationary point)" fra slides med penduleksemplet - bemærk, proceduren er ikke begrænset til et stationært (kritisk) punkt. Afhænger ligningssystemet af det kritiske punkt, og hvordan i så fald?
- Lav et plot af den identificerede ulineære funktion sammenholdt med dens linearisering (i et eller alle kritiske punkter).
- For hvert af de kritiske punkter, undersøg typen og stabiliteten heraf, og lav et faseplot i omegnen af det kritiske punkt.